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Anchored Conditionally-Gated Control for Wheeled Bipedal Balance and Disturbance Recovery

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ABSTRACT

Wheeled bipedal robots face a trade-off: the stiffness that buys precise standing shrinks the push-recovery envelope, while the integral action that removes equilibrium bias winds up during disturbances. The Anchored Conditionally-Gated Controller (ACC) answers both by *conditional activation*—a position integral confined to a 15 cm neighbourhood and enabled by an asymmetric envelope follower ($\tau_{\text{attack}} \approx 23$ ms, $\tau_{\text{release}} \approx 1.5$ s) that separates quiet stance from post-push ringdown. On a 10-DOF, 8.1 kg wheeled biped in MuJoCo (500 Hz physics, 100 Hz control) the architecture is dissected, not only demonstrated. A factorial ablation separates three coupled effects: gated damping produces the steadiness, the anchor integral the accuracy at a 3× cost in steadiness, and the gates are bit-identical to ACC in quiet stance yet decisive under disturbance. The push envelope is measurably chiral (−16.3 N), traced to the sign convention of the antisymmetric hip-roll/hip-yaw channels and closed by two control experiments: a mirror flips it, a sham mirror does not. A closed-form model predicts the 27.0 mm standing offset to 0.92 % across a fortyfold gain sweep, as the price of a bounded integrator memory. ACC holds 0.294±0.015 mm sagittal repeatability (N=10)—a 59-fold improvement over this ablation's proportional-only rung—recovers omnidirectional pushes (107 N median, 82 N worst case), stands 30 min at 0.0004 mm/min drift where six classical baselines and a full-state LQR all fall, and survives a 100 cm drop and 10 – 20 cm curbs. A 2 ms-resolved delay screen prices deployment at a 6 – 8 ms latency cliff that one retuned gain moves outward: a tuning artefact, not an architectural ceiling.

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1. Introduction

Wheeled biped robots need millimeter-level standing precision, omnidirectional push recovery, and terrain adaptation—but no single classical controller integrates all three: LQR handles balance (Klemm et al., 2019), WBC posture (Klemm et al., 2020), and MPC terrain (Wang et al., 2025); locomotion, separately again, through learned policies (Hwangbo et al., 2019).

The difficulty is fundamental. Lowering k_p widens the capture basin at the cost of balance stiffness; omitting integral action leaves a 1.3 Nm equilibrium bias that sustains a persistent, predominantly *sagittal* limit cycle regardless of k_p (N=10: 69.7 mm peak-to-peak sagittally against 13.4 mm laterally), on the one axis the wheels can drive; yet a global integral winds up during pushes.

We present the Anchored Conditionally-Gated Controller (ACC), which addresses both dilemmas through *conditional activation*. The core mechanism is a **proximity-gated anchor**: a position integral confined to a 15 cm neighborhood, enabled by an **asymmetric envelope follower** ($\tau_{\text{attack}} \approx 23$ ms, $\tau_{\text{release}} \approx 1.5$ s) that distinguishes quiet stance from post-push ringdown. Coupled with a **gated damping boost**, ACC achieves 0.294±0.015 mm repeatability on the sagittal axis under idealized conditions (clean sensors, 0 ms delay, N=10), at push performance equal to or better than a proportional-only baseline carrying no integral action at all. Absolute accuracy on that axis is a separate quantity: the robot parks 27.0 mm from the commanded home and then holds that standoff to 0.009 mm.

A factorial ablation under a single protocol (Table 1) separates three effects the architecture couples: quiet-stance *steadiness* comes from the gated *damping* terms rather than from the anchor integral, which buys *ac-*

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curacy instead; the proximity gate and the asymmetric envelope are provably inert during quiet stance, so only a displacement experiment can measure them; and the pitch safety gate is load-bearing with no disturbance applied. The gates are enabling conditions that make an otherwise inadmissible damping and integral configuration safe, not the proximate source of the precision. The paper is positioned within *structured classical priors* rather than learned control: the object of study is a hand-designed, interpretable torque assembly. Supplying the nominal base action of a residual-learning stack (Johannink et al., 2019; Silver et al., 2018) is the downstream use it was designed for, and it is future work; no residual policy is trained here and no claim below depends on one. Classical controllers are therefore the comparison of record: six of them—four 4-state TWIP variants and two 6-state coupled variants—all fall, and a full-state LQR linearized from the plant itself survives two orders of magnitude longer and still falls, so platform complexity, not model order or servo lag, dominates the failure. A preliminary model-free PPO (Schulman et al., 2017) run is retained as a *difficulty probe* and bounds nothing about what RL attains here.

The main contributions are:

(1) A **proximity-gated anchor** with asymmetric envelope follower achieving sub-millimeter idle precision without sacrificing omnidirectional push survival.
(2) A **two-channel conditionally-gated torque assembly** with flight recovery and per-leg terrain adaptation as independent extensions, each ablated at N=10.
(3) A **factorial ablation** (additive L0→L3 + subtractive S0→S5) against six classical baselines, plus two control experiments that test the comparison itself: a full-state LQR derived from the plant rather than a reduced model, and a sweep of the coupled model's one empirical constant.

2. Related Work

2.1. Wheeled Bipedal Platforms and Classical Control

Ascento (Klemm et al., 2019) demonstrated two-wheeled jumping with LQR balancing, later extended to WBC with kinematic loops (Klemm et al., 2020); SKATER (Wang et al., 2025) uses a hierarchical MPC whole-body controller with terrain adaptation; comparable platforms pair a balance controller with separate yaw, height and roll loops. Whleaper (Zhu et al., 2024) is closest to our morphology (10 DOF, three hip DOFs per leg); it applies LQR to balanced sliding and a separately trained PPO policy to walking, *switching* between the two by locomotion mode rather than composing them on one control output. Composing a classical prior with a learned residual on one output is the residual-RL construction (Johannink et al., 2019; Silver et al., 2018), applied recently to wheeled-biped balance (Zhang et al., 2025); ACC supplies that prior rather than replacing the learned part. A recent review (X. Liu et al., 2024) notes the absence of a unified classical solution.

Coupled pitch–roll LQR is standard on rigid two-wheeled platforms, where torque acts directly on the wheels and there are no articulated legs or leg-height dynamics. Our 6-state LQR reproduces that structure on a legged morphology and fails; the 4-state Direct Torque LQR, which removes the servo path, also fails, so actuation lag alone cannot account for the outcome (§5.4). These results are empirical, not a proof that LQR is structurally insufficient.

Cascade control and whole-body control via hierarchical inverse dynamics or task-priority QPs (Herzog et al., 2016) use fixed-gain nested or prioritized loops without conditional gates, and gain scheduling varies parameters continuously with one variable rather than responding to distinct physical regimes; the task-priority QP we benchmark standalone and as an ACC assist layer (§5.5) fails to resolve the precision–recovery trade-off for that reason. Split-range, selector and override architectures (Åström & Hägglund, 2006) already blend continuously in process control; ACC's contribution is the synthesis of physically-motivated gating surfaces—spatial proximity, temporal velocity envelope, pitch safety interlock—derived from wheeled-biped failure modes rather than from generic process-variable limits.

2.2. Disturbance Rejection, Anti-Windup, and ACC's Distinction

DOB (Ohnishi et al., 1996) and ADRC (Han, 2009) provide temporal disturbance estimation but cannot distinguish a push from post-push ringdown—both occupy the same frequency band. Classical anti-windup—back-calculation, conditional integration and the unified observer-based framework (Åström & Rundqwist, 1989)—triggers on *actuator saturation*; ACC's anchor instead uses **spatial, physics-conditioned gating**, confining integration to a 15 cm neighborhood and distinguishing quiet stance from ringdown by a velocity envelope, which replaces the estimation-speed-versus-noise trade-off with a time-domain attack-versus-release trade-off. Push recovery via capture point (Englsberger et al., 2015) and flight-phase control (Roscia et al., 2023) is complementary, and terrain adaptation is usually exteroceptive or learned (Lee et al., 2020); ACC extends proprioceptive contact-point estimation to wheeled bipeds. Hyon & Cheng (2006) made a humanoid passive to unmeasured external forces by gravity compensation with optimal contact-force distribution; ACC meets the same requirement by switching regime on proprioceptive gating variables rather than redistributing contact forces.

3. Robot Platform

3.1. Robot Morphology and Simulation

The robot (Fig. 1) is a ten-DOF wheeled biped with five actuated joints per leg: hip roll (HR), hip yaw (HY), hip pitch (HP), knee (KN), and drive wheel (WH). Mass: 8.1 kg; thigh: 0.26 m; shin: 0.28 m; wheel radius: 0.06 m; hip width: 0.23 m. Torque limits: 30 Nm (HR, HY, WH), 150 Nm (HP, KN). MuJoCo (Todorov et al., 2012) simulation runs at 500 Hz ($\Delta t_{\text{sim}} = 0.002$ s); ACC and every classical baseline run at 100 Hz ($\Delta t_{\text{ctrl}} = 0.01$ s, 5 physics substeps), with the baselines additionally measured at their 50 Hz design rate and the PPO reference reported at the 50 Hz rate it was trained at. All experiments use a 400 Nm/s torque rate limit and raw torque commands direct to actuators. Integrator: implicitfast; cone: pyramidal (elliptic for wheels, μ_s=1.2, μ_k=0.8 body); armature 0.008 kg·m^2 (wheels) and 0.02 kg·m^2 (leg joints); contact `solref={0.002,1}` (body), {0.005,0.7} (wheels). Softening `solref[0]` to 0.02 (tire-on-asphalt) increases planar idle RMS by ~60 % to ~1.2 mm, so every sub-millimeter idle figure is conditional on stiff ground contact.

Height convention

CoM-z, the whole-body centre-of-mass height above ground, is what ACC commands and regulates; the nominal standing pose is h_{CoM}=0.404 m. **Base-z**, the torso root-body height (`qpos[2]`), sits 13.2 cm above it in that pose, varying by less than 1 cm over the postures used here. All ACC results are in CoM-z; the PPO reference (§5.4) and the WBC baselines (§5.5) command base-z and are labelled accordingly, since substituting one convention for the other injects the full 13.2 cm offset as a reference error. ACC's posture map is calibrated at 0.354 m and 0.454 m and interpolated between them; outside that band joint targets are linearly extrapolated and bounded to $[0.254, 0.554]$ m, and it is that ±10 cm bound the per-leg height split (§4.6) operates against.

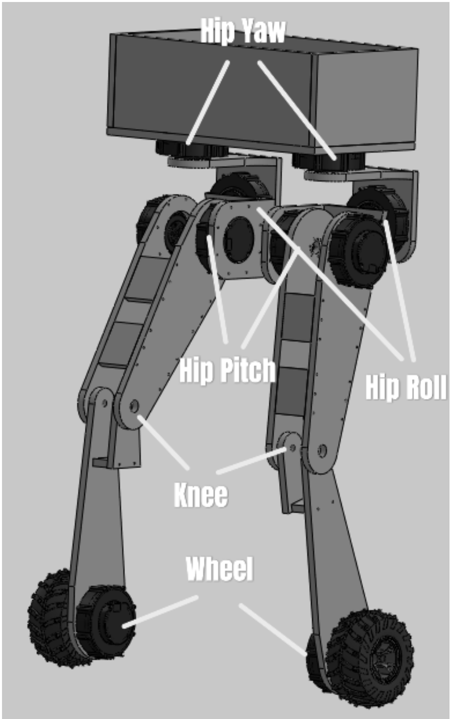

Figure 1. Robot joint layout with world frame (right-handed, Z up). Both wheel axles lie along world X and the wheel links are separated by 0.271 m along it, so the rolling—and therefore sagittal—direction is world Y and world X is the lateral (track) axis; all axis-resolved results follow that convention (Sec. 5.1). Each leg: hip roll (HR), hip yaw (HY), hip pitch (HP), knee (KN), wheel (WH).

4. ACC Formulation

4.1. Two-Channel Torque Assembly

ACC assembles a PD balance core, a proximity-gated anchor integral and posture regulation into two torque channels—wheel torque ($\boldsymbol{\tau}_w \in \mathbb{R}^2$) and leg-joint torque ($\boldsymbol{\tau}_q \in \mathbb{R}^8$)—with flight and terrain as independent extensions (Fig. 2):

$$\boldsymbol{\tau}_w = \boldsymbol{\tau}_{\text{balance}} + g_{\text{anchor}}\boldsymbol{\tau}_{\text{anchor}} + g_{\text{flight}}\boldsymbol{\tau}_{\text{flight}},$$
$$\boldsymbol{\tau}_q = \boldsymbol{\tau}_{\text{posture}}(h^{\text{cmd}}) + g_{\text{terrain}}\Delta\boldsymbol{\tau}_{\text{posture}}(h^{\text{cmd},L},\ h^{\text{cmd},R}). \quad (1)$$

Every gate g_* except the flight gate is a *smoothstep* function of a physical condition (proximity, quietness, attitude, terrain disparity) rather than a discrete switch:

$$\mathrm{ss}(x; a, b) = 3t^2 - 2t^3, \qquad t = \mathrm{clip}\left(\frac{x-a}{b-a}, 0, 1\right). \quad (2)$$

Equation (2) is the standard Hermite blend, C^1 in x with zero derivative at both knees. Continuity matters concretely: a hard threshold injects a torque step equal to the full gated component, which at 100 Hz under a 400 Nm/s rate limit reaches the plant as an unmodelled impulse. The one exception is g_{flight}, a discrete load-based hysteresis with asymmetric engage/release timers, because contact loss is a binary physical event.

Wheel torques control sagittal balance and yaw; leg-joint torques control posture and per-leg height adaptation. The two channels are coupled only through the plant—no term in $\boldsymbol{\tau}_w$ reads a leg-posture state and none in $\boldsymbol{\tau}_q$ reads the sagittal balance state—which is what allows flight recovery and terrain adaptation to be added as independent extensions without retuning the anchor. Unlike the industrial override architectures of §2.2, which select *among parallel controllers* on process-variable limits, ACC gates *components within a single torque assembly* on physical quantities of three kinds—spatial, temporal and safety-interlock—each chosen because a specific measured failure mode of this platform is expressible in it.

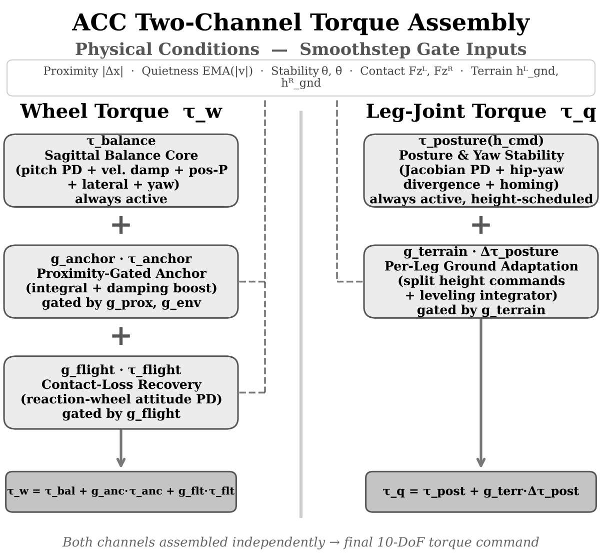

Figure 2. ACC two-channel torque assembly. Physical conditions gate each component via continuous smoothstep functions (dashed).

4.2. *Sagittal Balance Core ($\tau_{balance}$)*

The sagittal balance component is always active and acts entirely in the wheel-torque channel:

$$\boldsymbol{\tau}_{\text{balance}} = \begin{bmatrix} \tau_{\text{com}} + \tau_{\text{diff}} \\ \tau_{\text{com}} - \tau_{\text{diff}} \end{bmatrix}, \quad \begin{array}{l} \tau_{\text{com}} = \tau_{\text{pitch}} + \tau_{\text{damping}} + \tau_{\text{pos}}, \\ \tau_{\text{diff}} = \tau_{\text{lat}} + \tau_{\text{yaw}}, \end{array} \quad (3)$$

with common-mode and differential components

$$\tau_{\text{pitch}} = k_p \cdot \theta + k_d \cdot \dot{\theta}, \quad (4)$$

$$\tau_{\text{damping}} = -k_{\text{vel}} \cdot v_{\text{sag}}, \quad (5)$$

$$\tau_{\text{pos}} = -k_{\text{pos}} \cdot \mathrm{clip}(\Delta x, \pm 0.10\,\mathrm{m}), \quad (6)$$

$$\tau_{\text{lat}} = k_{p,\text{roll}} \cdot \phi + k_{d,\text{roll}} \cdot \dot{\phi}, \quad (7)$$

$$\tau_{\text{yaw}} = k_{p,\text{yaw}} \cdot e_{\psi} + k_{d,\text{yaw}} \cdot \omega_z, \quad (8)$$

where θ is torso pitch, v_{sag} the sagittal body velocity, Δx the sagittal position error from the latched home position, ϕ the roll angle and e_{ψ} the yaw error.

The P position term (k_{pos}, clipped to ± 0.10 m, saturating at ± 4 Nm) keeps the robot within $\pm 4 - 6$ cm of home but leaves a 1.3 Nm equilibrium bias sustaining a perpetual limit cycle, which is what motivates the anchor integral. Saturation-triggered anti-windup does not address it, because the critical failure is *state-dependent*: during a push, position error grows to 10 – 30 cm while the wheels remain within their 20 rad/s range, so a saturation trigger never fires and ACC uses spatial confinement instead. Pitch stiffness is fixed at k_p=50; a scheduled variant softening it to 35 during pushes was tested and rejected (S0 vs. S1, Table 3).

4.3. *Proximity-Gated Anchor ($g_{anchor}\tau_{anchor}$)*

The anchor mechanism is the core contribution. It adds two terms to the wheel-torque channel that are active only when the robot is near home *and* genuinely at rest:

$$\begin{aligned} \tau_{\text{anchor}} = K_i \int g_{\text{prox}} \cdot g_{\text{env}} \cdot (-\Delta x)\, dt \\ + g_{\text{boost}} \cdot K_{\text{boost}} \cdot (-v_{\text{sag}}), \end{aligned} \quad (9)$$

where the gates are continuous smoothstep functions equal to 1 when the robot is "at home and quiet" and transitioning smoothly to 0 otherwise:

$$g_{\rm prox}(\Delta x) = 1 - \mathrm{ss}\big(|\Delta x|;\ 0.05\,\mathrm{m},\ 0.15\,\mathrm{m}\big), \tag{10}$$

$$g_{\rm env}(v_{\rm sag}) = 1 - \mathrm{ss}\big(\mathrm{EMA}(|v_{\rm sag}|);\ 0.25\,\mathrm{m/s},\ 0.50\,\mathrm{m/s}\big), \tag{11}$$

$$g_\theta(\theta, \dot\theta) = \big[1 - \mathrm{ss}(|\theta|;\ 2^\circ,\ 12^\circ)\big] \times \big[1 - \mathrm{ss}(|\dot\theta|;\ 2^\circ/\mathrm{s},\ 15^\circ/\mathrm{s})\big], \tag{12}$$

$$g_{\rm boost} = g_{\rm prox} \cdot g_{\rm env} \cdot g_\theta. \tag{13}$$

Each gate is dimensionless, monotone and C^1, and three physically distinct conditions must hold simultaneously for the anchor to integrate: spatially near home ($g_{\rm prox}$), temporally quiet ($g_{\rm env}$) and attitudinally settled (g_θ). The pitch gate closes when either pitch or pitch rate leaves its band, preventing the damping boost from fighting a gentle sustained push that does not trigger the velocity envelope.

Asymmetric envelope follower. Reliable quiet-stance detection rests on an asymmetric exponentially-weighted moving average (EMA) of $|v_{\rm sag}|$:

$$\mathrm{EMA}_t = \begin{cases} \alpha_a|v_t| + (1-\alpha_a)\mathrm{EMA}_{t-1}, & |v_t| > \mathrm{EMA}_{t-1}, \\ \alpha_r|v_t| + (1-\alpha_r)\mathrm{EMA}_{t-1}, & \text{else}, \end{cases} \tag{14}$$

with time constants $\tau_{\rm attack}{\approx}23$ ms and $\tau_{\rm release}{\approx}1.5$ s as the primary specification, the discrete coefficients following as $\alpha_a{=}1{-}e^{-\Delta t/\tau_a} \approx 0.35$ and $\alpha_r \approx 0.007$ at 100 Hz. The asymmetry is essential: a symmetric EMA is *bistable*—post-push oscillation holds it in a middle band where the gate is half-open and the boost cannot collapse the cycle, as the symmetric-EMA ablation of §5 confirms—whereas the fast-attack/slow-release envelope rises to 1 only when oscillation truly decays and drops to 0 within ~25 ms of a disturbance. The mechanism is a peak-hold envelope detector; ACC's contribution is coupling it to a spatial proximity gate for integral conditioning. Since $\mathrm{EMA}_0{=}0$ biases $g_{\rm env}{\approx}1$ for ~7.5 s at startup, all measurements discard a 3 – 5 s settle interval.

4.4. Posture and Yaw Stability ($\tau_{posture}$)

Posture regulation uses Jacobian-based PD gains scheduled across 23 heights:

$$\begin{aligned}\boldsymbol{\tau}_{\rm posture} = \boldsymbol{\tau}^{\rm jac}_{\rm PD}(q, \dot q, h^{\rm cmd}) + \boldsymbol{\tau}_{\rm div}(q^L_{\rm hy}, q^R_{\rm hy}) \\ + g_{\rm settled}\,\boldsymbol{\tau}_{\rm homing}(q - q_{\rm ref}),\end{aligned} \tag{15}$$

where $\boldsymbol{\tau}_{\rm div}$ damps hip-yaw divergence and $\boldsymbol{\tau}_{\rm homing}$ restores nominal posture when upright and settled. Yaw return uses wheel-differential torque.

4.5. Contact-Loss Recovery ($g_{flight} \cdot \tau_{flight}$)

When both wheels lose ground contact, ACC detects this from per-wheel normal forces and substitutes a flight attitude controller using the wheels as reaction flywheels:

$$g_{\rm flight} = \begin{cases} 1, & \text{if } F^L_z, F^R_z < 0.5mg \text{ for } {\ge}2 \text{ steps}, \\ 0, & \text{if } \min(F^L_z, F^R_z) \ge 0.5mg \text{ for } {\ge}5 \text{ steps}, \end{cases} \tag{16}$$

$$\tau_{\rm flight} = \mathrm{clip}\big(2.0\theta + 0.5\dot\theta - 0.02\omega_{\rm wheel},\ \pm 2\,\mathrm{Nm}\big). \tag{17}$$

Forward wheel spin pitches the torso nose-up during descent, and the 5-step release hysteresis prevents balance-torque re-engagement during bounce. The wheel–inverted-pendulum loop this override replaces is itself a reaction-flywheel controller once the wheels are unloaded, so the override's measured contribution is to bound landing-attitude variance near the upper end of the drop envelope rather than to enable recovery (§5.3).

4.6. Per-Leg Ground Adaptation ($g_{terrain}$)

When the two wheels rest at different ground heights—a curb straddle, an oblique ledge—a single torso height command forces the torso to roll by the step angle. ACC instead maintains an *independent ground estimate per leg* and issues two height commands. The estimate is taken from the wheel contact-point height, which is exact at any body tilt for a rigid wheel, and is updated only while that wheel carries real load: $\hat h^i_{\rm gnd} \leftarrow \hat h^i_{\rm gnd} + \alpha_g(z^i_{\rm contact} - \hat h^i_{\rm gnd})$ for $F^i_z \ge 0.2mg$, $i \in \{L, R\}$, with $\alpha_g = 0.35$ per control step to reject contact chatter. The $0.2mg$ load test rather than mere geometric touching is deliberate: a lightly touching wheel in the act of unloading injected a 2.9 cm phantom ground step on the 15 cm curb. An unloaded wheel freezes its estimate, the one exception being the unambiguous signal that its ground is gone—the wheel centre descending *below* its own believed ground. Writing $\delta^i = \hat h^i_{\rm gnd} - z^i_{\rm contact}$, a $\delta^i >$ 3 mm drives the estimate down at $\dot{\hat h}^i_{\rm gnd} = -(v_0 + k_s\delta^i)$ with $v_0 = 0.5$ m/s, $k_s = 50\,\mathrm{s}^{-1}$. The proportional term matters: a fixed slew converged in 0.30 s, by which time a 20 cm dismount had already tipped the robot past recovery, whereas this law converges within two control steps, so the leg extends *during* the fall. When both wheels are airborne both estimates track the mean wheel height.

The step $\Delta h = \hat{h}^L_{\text{gnd}} - \hat{h}^R_{\text{gnd}}$ is then absorbed by the legs rather than the torso. The split is kinematic, not a linear height offset: with $L(\cdot)$ the forward-kinematic leg length and L^{-1} its inverse over the calibrated posture range, the leg on the higher ground is shortened by the full step,

$$L_{\text{high}} = L\left(h^{\text{cmd}}\right) - |\Delta h|, \tag{18}$$

$$h^{\text{high}} = L^{-1}\left(L_{\text{high}}\right), \qquad h^{\text{low}} = h^{\text{cmd}}, \tag{19}$$

both clamped to the calibrated envelope extended by 10 cm at each end. To first order this is the familiar $h^{\text{cmd}} \mp \Delta h/2$ split, but the $h \mapsto L$ map is markedly non-linear near full squat, the linear proxy erring by 12 cm at the lower extrapolation limit. If L_{high} falls below the shortest reachable leg, ACC raises the *whole* robot until the short leg fits and reports the remainder as an expected torso roll $\arctan(-\varepsilon/w_{\text{track}})$ (ε the unspanned step, $w_{\text{track}} = 0.278$ m), which keeps a legitimate straddle from being read as an incipient fall. Because the $h \mapsto L$ table is itself extrapolated at the extremes, a purely feedforward split left a systematic 6° torso roll on the 10 cm curb; a closed-loop levelling integrator trims it by the *measured* roll at 0.15 m/(rad s) with ±6 cm of authority, integrating only when both wheels are loaded, the step is genuine ($|\Delta h| \geq 2$ cm) and the estimate is settling (<0.15 m/s); the first condition keeps a 180° turn from winding in an 8.6° bias, the second keeps the transient of a curb climb out of the levelling error.

The expected roll must also reach the *gates*. Heading hold is a differential wheel torque scaled by a stability envelope whose roll factor decays over 1–5°, so the quasi-static 2.5–4.0° of a 20 cm straddle read as an incipient fall and suppressed heading hold for the whole crossing. ACC therefore scales the roll band with the commanded split and restores both the differential and the mean-ground height reference once the straddle has *settled*, decided by the split rate against a 0.2 s average ($|\Delta\dot{h}|$ is 0.05–0.13 m/s while a wheel climbs, below 0.016 m/s astride). Relaxing any earlier keeps drift authority alive over an edge and topples the robot on a 45° oblique exit; on flat ground $\Delta h \equiv 0$ and every term vanishes identically.

4.7. Complete Numerical Parameter Set

The core gains are k_p=50 Nm/rad, k_d=3 Nm/(rad/s), k_{pos}=40 Nm/m (clipped at ±4 Nm), k_{vel}=15 Nm/(m/s) at damping scale 1.5, k_{roll}=55 Nm/rad, k_{yaw}=1.0 Nm/rad; the anchor uses K_i=4.0 Nm/(m s) capped at 2 Nm with a 5×10^{-3} per-step leak, boost scale 5.0, g_{prox} band 0.05–0.15 m, g_{env} band 0.25–0.50 m/s, g_{stab} bands 2–12° and 2–15 ° s^{-1}; flight uses P/D = 2.0/0.5 with 0.02 Nm/(rad/s) wheel damping and 2/5-step engage/release at F_z=0.5mg (0.2mg for terrain). The full set (50+ gains) is in the supplementary material.

5. Experimental Validation

ACC and all six classical baselines are evaluated at 500 Hz physics / 100 Hz control under a 400 Nm/s per-joint torque rate limit, so the headline comparison is rate-matched by construction; the PPO reference is reported at the 50 Hz rate its policy was trained at. Each classical baseline was additionally measured at its own 50 Hz design rate: the fall rate is 100 % in all twelve cells and five of the six survive *longer* at 50 Hz, so reporting them at 100 Hz is the rate-matched and not the flattering choice. Seeds are deterministic: 20260727+trial_id for idle standing, 20260728×100+rep for the push ablation, and {0, 42, 123} for the classical baselines.

5.1. Anchored Standing

5.1.1. Metrics and axis convention

"Standing precision" names two quantities that are not interchangeable. With $\mathbf{p}(t) \in \mathbb{R}^2$ the horizontal base position over a window of length T, $\mathbf{p}_0$ the commanded home and $\bar{\mathbf{p}}$ the trial mean,

$$A = \left(\frac{1}{T}\int_0^T \|\mathbf{p}(t) - \mathbf{p}_0\|^2\, dt\right)^{1/2}, \tag{20}$$

$$R = \left(\frac{1}{T}\int_0^T \|\mathbf{p}(t) - \bar{\mathbf{p}}\|^2\, dt\right)^{1/2} \tag{21}$$

define the *accuracy* A, which retains any DC parking offset, and the *repeatability* R, which removes it. The two differ by more than an order of magnitude on this platform, so both are reported for every configuration. Each is resolved along the sagittal axis, the lateral axis and the full plane, because the two horizontal axes are actuated by different means: the **sagittal** axis is world y, the rolling direction and the only one on which wheel torque produces ground force, while the **lateral** axis is world x, held passively by the wheel contacts and the roll restoring moment of the track width. A dynamic check confirms the assignment (`scripts/verify_body_axes.py`): a 90 N impulse along y is absorbed at a peak wheel speed of 52 rad/s leaving 4.8 mm of residual offset, whereas the same impulse along x leaves a permanent 23.2 mm offset. The anchor integral and the gated damping boost both live in the wheel-torque channel, so both act on the sagittal axis only.

5.1.2. Protocol

Every row of Table 1 was re-measured under one protocol: 25 s from the nominal standing pose with randomised initial conditions (σ=0.005 rad per joint, clipped to limits; σ=1 mm on root height), with A, R taken relative to each trial's own t=0 pose after a settle long enough to cover the EMA$_0$=0 transient to within 1 %. A trial is a fall if $|\theta|>46°$ or root height <0.15 m; confidence intervals are Student-t. The protocol measures the ability to *stay* put; coming back is measured in Sec. 5.2.

5.1.3. Result

ACC holds **sub-millimetre sagittal repeatability**, R_{sag} = 0.294±0.015 mm (95% CI [0.283, 0.305] mm) across ten randomised trials with no falls—the figure of merit for the platform, since sagittal is the actuated inverted-pendulum axis. Precision is assessed from the relative spread *between* configurations under an identical protocol rather than against a simulator noise floor: removing a single mechanism (S2) moves R_{sag} sixteenfold with non-overlapping SDs. Against the P-only baseline L0, which sustains a 69.7 mm peak-to-peak sagittal limit cycle, the like-for-like improvements are **59.4**× sagittal, **22.6**× planar and **5.0**× lateral—the largest gain on the hard axis and the smallest on the easy one, the correct signature for a wheel-torque controller, since every term ACC adds beyond L0 produces ground force along the rolling direction and none can produce lateral force at all.

Accuracy and repeatability are distinct on this axis. ACC parks at $\bar{s}$ = −27.0 mm from commanded home and holds that standoff to 0.009 mm SD across trials, so planar accuracy is A_{xy} = 26.994 mm, a 1.8× improvement over L0 against the 59× gain in repeatability. Its magnitude is fully accounted for: instrumenting the sagittal torque balance predicts the offset in closed form from a *leaky* anchor integral whose ∼2 s forgetting horizon fixes its DC gain at the finite $k_{I,\text{dc}}$=7.96 Nm/m—so a constant load leaves a finite error by construction—and a 1.44° error in the feedforward pitch reference, giving 24.88 mm against 24.99 mm measured and holding across a 40× gain sweep and a five-point height sweep to within 0.92 %. The leak also buys the steadiness: a 10× smaller λ cuts the offset to −16.31 mm but degrades window jitter twelvefold. Table 1 publishes a deliberately chosen point on that trade-off.

5.1.4. The measurement window is a protocol choice, not a stability horizon

Every row of Table 1 is measured over 20 s, and Section 5.4.1 declines to promote a competing design specifically because a 20 s horizon flatters it. That standard has to cut both ways, so it is applied here. Re-running the ACC row—identical seeds, initial conditions, profile and metrics, with only the window lengthened—gives Table 2. The 20 s column reproduces the published 0.294±0.015 mm exactly, which fixes the two runs as the same measurement rather than a re-tuned one.

The 20 s figure is therefore the *conservative* one: that window still contains the tail of the settle, so lengthening it improves repeatability (3.3× at 300 s, 7.9× at 1800 s) rather than degrading it. What the settle terminates in is a fixed point, not a slow mode: after ≈35 s the sagittal coordinate is constant to 2×10^{-6} mm per 30 s block for the remaining twenty-nine minutes, the parking offset holds at −26.763 mm, CoM height holds 0.4068 m, and the only motion left anywhere in the state is a lateral creep of 0.14 mm/hour on the passively held axis. The separation from the full-state LQR of Section 5.4.1 is thus qualitative rather than one of degree—that design accumulates 0.43 – 0.63 m of planar drift and falls—because ACC's position error is referenced to a latched home by two explicit terms, the clipped proportional τ_{pos} of Eq. (3) and the anchor integral on $-\Delta x$ of Eq. (9). What Table 1 reports is a window chosen to match the baselines, not the longest interval over which the claim holds.

5.1.5. Which mechanism produces the precision

The lower blocks of Table 1 isolate the contributions, and three of the four findings contradict the attribution that the gate structure invites.

(i) Idle steadiness comes from the damping ladder; the integral buys accuracy instead. M1 disables the anchor integral (K_i=0) with every gate in place, and its *repeatability* is slightly better than ACC's while its *accuracy* is worse by 4.2 mm; conversely M2 keeps the integral and every gate but reverts the two velocity-damping coefficients to their L2 values, reproducing the L2/L2.5 limit cycle exactly. Steadiness is therefore attributable to the damping terms in two stages: raising the velocity-damping scale takes R_{sag} from the L2 limit cycle to 3.97 mm (M3) and the gated boost takes it the rest of the way, a combined factor of 86. *(ii) The damping boost acts sagittally, as it must.* S2 degrades

Table 1. Idle standing precision across the full ablation ladder. All rows re-measured under the single protocol of Sec. 5.1; values are mean±SD over N=10 trials.

	Configuration	R_{sag} (mm) sagittal rep.	R_{xy} (mm) planar rep.	R_{lat} (mm) lateral rep.	$\bar{s}$ (mm) sag. offset	Survival
Additive ladder (bottom-up)						
L0	P-only (PD, no position, no integral)	17.461 ± 0.258	17.845 ± 0.279	3.675 ± 0.246	−44.8	10/10
L1	+ position homing (±0.10 m)	20.397 ± 0.017	20.414 ± 0.019	0.834 ± 0.066	−30.4	10/10
L2	+ ungated integral K_i	25.026 ± 0.039	25.056 ± 0.043	1.212 ± 0.110	−27.5	10/10
L2.5	+ proximity gate only (no envelope, no boost)	25.198 ± 0.017	25.227 ± 0.020	1.205 ± 0.099	−27.6	10/10
Proposed controller						
L3/S1	**ACC** (full anchor, fixed k_p=50)	**0.294 ± 0.015**	**0.788 ± 0.069**	**0.730 ± 0.073**	**−27.0**	**10/10**
Subtractive ablation (top-down from ACC)						
S2	− gated damping boost	4.772 ± 0.069	4.824 ± 0.067	0.703 ± 0.069	−26.8	10/10
S3	− asymmetric envelope (symmetric EMA)	0.294 ± 0.015	0.788 ± 0.069	0.730 ± 0.073	−27.0	10/10
S4	− proximity gate (integral always on)	0.294 ± 0.015	0.788 ± 0.069	0.730 ± 0.073	−27.0	10/10
S5	− pitch safety gate g_θ	—	—	—	—	**0/10**
Single-mechanism isolation (ACC with one term disabled)						
M1	K_i=0 (anchor integral removed, gates intact)	0.089 ± 0.035	0.733 ± 0.072	0.727 ± 0.071	−31.2	10/10
M2	damping terms reverted to their L2 values	25.198 ± 0.017	25.227 ± 0.020	1.205 ± 0.099	−27.6	10/10
M3	K_i=0 *and* boost = 0	3.973 ± 0.130	4.037 ± 0.128	0.710 ± 0.069	−31.1	10/10

100 Hz control, 20 s window after a 5 s settle, clean sensors, 0 ms delay. R = repeatability (Eq. (21)); $\bar{s}$ = mean signed sagittal parking offset; accuracy on an axis is that offset and the repeatability in quadrature, so the A columns are omitted (the identity holds to 0.003 mm on every surviving row). Sagittal is world y, lateral world x; the anchor integral and the damping boost act sagittally only, so R_{sag} and $\bar{s}$ discriminate between configurations while R_{lat} measures a passively held axis. Produced by `scripts/collect_idle_ladder.py`; raw output `outputs/paper_verification/idle_ladder.json`.

Table 2. Quiet stance at three window lengths. Same seeds, initial conditions, profile (`V3_ANCHOR`) and metric definitions as Table 1; only the record length differs. R_{sag} is over the full window; drift is a least-squares rate over the same span.

Window	N	No fall	R_{sag} (mm)	Drift_{sag} (mm/min)
20 s (Table 1)	10	10/10	0.294(15)	—
300 s	10	10/10	0.089(5)	+0.0149
1800 s	4	4/4	0.037(2)	+0.0004

100 Hz control, 500 Hz physics, 5 s settle before the window, clean sensors, 0 ms delay, seeds 20260727+i. Parenthesised digits are the SD in the last place. Produced by `scripts/idle_longhorizon.py`; raw output `outputs/paper_verification/idle_longhorizon_{300,1800}s.json`.

sagittal repeatability sixteenfold while leaving lateral repeatability statistically unchanged (overlapping CIs), the only outcome consistent with a boost inside the wheel-torque channel.

(iii) The proximity gate and the asymmetric envelope are inert during quiet stance. Ablating either (S4, S3) is *bit-identical* to ACC on every idle metric—the base never leaves the 5 cm inner band and the velocity envelope never rises into the attack branch—so both are correctly evaluated only in Table 3. *(iv) The pitch safety gate is required even to stand.* S5 falls in 10/10 trials during quiet stance with no push applied: removing g_θ leaves the damping boost free to act on the pitch state at large excursions, and the resulting positive feedback diverges from the ±0.005 rad initial perturbation alone. The anchor's *gates* are therefore what make an aggressive damping and integral configuration admissible without windup or divergence, while the damping terms they admit are what deliver the steadiness.

5.2. Omnidirectional Push Recovery

Push recovery is evaluated by polar sweep: 10 – 160 N impulses of 7 control steps applied at the torso in 8 world-frame directions (θ= ± 90° sagittal, θ=0°/180° lateral), each threshold located by binary search to 5 N, with survival defined as torso pitch < 46° for 17 s post-push. F_{min} is the maximum magnitude survived in *every* direction and F_{med} the median.

The envelope is anisotropic by construction. The two lateral bearings average 98.2 N against 131.3 N for the two sagittal ones, and the weakest bearing of all is the oblique 45° (82.3 N, 23 % below F_{med}). The asymmetry follows from the morphology rather than from tuning: the wheels can drive the contact patch back under the CoM sagittally but produce no lateral ground force, so raising the lateral bearing requires a centralised lateral task (§5.5), not a gain change.

The envelope is also chiral, and the chirality is in the controller. Superposed on the anisotropy is a left–right asymmetry: 135° survives 40.3 N more than −135° (Fig. 3). The plant cannot produce it—mirroring the model about the sagittal plane reproduces its mass, inertia and geometry to 0.05 mm and 1×10^{-7}—and a harmonic fit localises the violation in the mirror-forbidden term ($b_2 = -16.3$ N against $b_1 = 0.3$ N). The source is the sign convention of the two differential channels: under a mirrored state the commanded torque maps to its own mirror image exactly on hip-pitch, knee and wheel and fails only on hip-roll and hip-yaw, both assembled antisymmetrically—even rather than odd under the mirror map. Mirroring those channels flips b_2 to 15.6 N at unchanged magnitude, while a sham mirror leaves it at −15.7 N. We report the effect without adopting the correction, since it redistributes the envelope without enlarging it ($F_{\min}$ 82.5→79.7 N). Later protocols apply a single impulse along world $+x$, a lateral and below-median bearing (95.6 N), so those experiments are conservative probes.

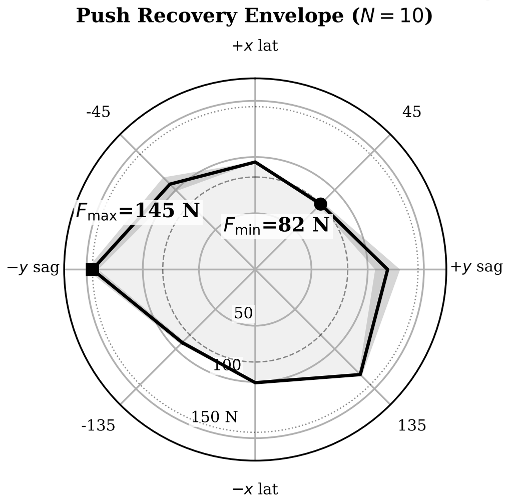

Figure 3. Omnidirectional push envelope of ACC—the *same* run as row S1 of Table 3. Solid curve: per-bearing mean survivable push over N=10 repetitions; shaded band ±1 SD; radial axis linear in force from 0 N. The envelope spans 82.3 – 144.8 N and is both anisotropic and chiral.

Table 3. Factorial ablation of ACC components: push-recovery and ringdown metrics. Clean sensors, all rows N=10, mean±std.

	Configuration	$F_{\min}$ (N)	F_{med} (N)	Ringdown
Additive (bottom-up)				
L0	P-only (PD, no pos., no I)	70.0±1.0	122.1±2.1	never
L1	+ Pos. homing (±0.10 m)	74.8±0.8	130.5±1.2	never
L2	+ Integral K_i (no gate)	74.8±1.0	131.1±1.1	—
L2.5	+ Prox. gate only (no EMA, no boost)	75.3±0.8	131.9±2.4	—
L3	+ Full anchor (gate, EMA, boost)	**82.3±2.2**	106.7±7.0	**9.0** s
Subtractive (top-down from ACC)				
S0	ACC + scheduled k_p (rejected)	81.0±2.2	109.2±7.6	9.0 s
S1	**ACC** (canonical, fixed k_p=50)	**82.3±2.2**	106.7±7.0	**9.0** s
S2	−Damping boost	80.8±0.9	116.2±5.3	never
S3	−Asym. EMA (sym. EMA)	81.6±1.8	106.2±6.7	never
S4	−Prox. gate (global I)	80.5±2.7	109.2±7.6	windup
S5	−Pitch gate g_θ	<10	<10	never

$F_{\min}$ and F_{med} are the minimum and median survivable push over 8 bearings, each located by binary search to 5 N tolerance; idle metrics for the same configurations are in Table 1. The S5 entries are left-censored at the search floor: in all 80 trials the bisection never located a survivable magnitude in [10 N, 160 N]. Welch's unpaired two-sided t-test (N=10 per group, Bonferroni-adjusted α=0.007): L0→L1 $p < 1 \times 10^{-9}$, d=5.4; L1→L2 p=0.88; L2→L2.5 p=0.19; L2.5→L3 $p < 2 \times 10^{-6}$, d=4.1. No subtractive row differs significantly from ACC except S5 ($p < 1 \times 10^{-14}$). At N=10 the 95% CI half-width is ≈0.72×std, so sub-2 N differences are not resolvable. Each trial starts from a freshly constructed controller context. Produced by `scripts/replicate_ablation_n10.py`; raw output `outputs/paper_statistics/`.

Table 3 reports the factorial ablation. Additively, only homing (L1) and the full anchor (L3) add significant margin; the global integral and the proximity gate contribute nothing to push *margin* on their own, consistent with the gate's role being windup prevention rather than capture-basin enlargement, and the full anchor is simultaneously the only configuration in the ladder that settles. Subtractively the striking result is how flat the ablation is: every single-mechanism removal except the pitch gate leaves $F_{\min}$ within 1.8 N of ACC and none is significant, so capture basin is set by the anchor as a whole rather than by any one of its four gates—including gain scheduling (k_p=50 fixed costs nothing measurable, S0) and the damping boost, which dominates idle precision yet leaves push margin indistinguishable from ACC's (S2). The one mechanism the push data *do* isolate is the pitch gate: removing it (S5) leaves the robot unable to survive even the search floor. Post-push ringdown decays within 9 s under ACC, whereas P-only oscillates indefinitely because the 1.3 Nm equilibrium bias sustains the limit cycle; the symmetric-EMA ablation exhibits the bistable failure the asymmetry is designed to prevent, and the ungated integral winds up during the push and prevents settling entirely.

5.3. *Flight Recovery and Terrain Adaptation*

Both extensions run at the same rate limit and control rate as the core with no re-tuning of the anchor. Table 4 reports both.

Table 4. Flight recovery and per-leg terrain adaptation, as configuration × scenario ablation matrices. Cells are mean±std over N=10 randomised-IC trials; a count in parentheses marks the trials that met the success criterion where it is not 10/10.

Contact-loss recovery — peak $|\theta|$ (°)

Configuration	Drop 100 cm	Drop 60 cm	Ledge 50 cm	Ledge 20 cm
ACC (full)	24.0±1.2	18.2±1.0	27.1±2.0	19.6±0.3
−Attitude PD[a]	27.8±0.3 (9/10)	—	31.6±2.2	—
−Override	13.0±0.5	—	10.2±0.4	—

Per-leg terrain — straddle roll $|\phi|$ (°), curb at 0.15 m/s

Configuration	10 cm	15 cm	20 cm
ACC (full)	2.4±0.1	2.9±0.2	3.9±0.1
−Straddle gating	—	—	14.9±0.1 (0/10)
−Levelling trim[b]	4.5±2.3	10.7 (1/10)	fall (0/10)
−Height split	fall (0/10)	fall (0/10)	fall (0/10)

Success is *settled* for drops and ledges (flat pitch/roll window, tail height error ≤12 mm, residual speed ≤0.05 m/s) and *traversed* for the curb—the elevated wheel stays on the 36 cm slab for the whole 2 m, which a survival criterion cannot distinguish from yawing off the side halfway along. Straddle roll is taken over traversing trials only. At the envelope edge—a drop of 150 cm, N=5—peak pitch is 34.0±2.1 with the override against 53.6±35.0 without it, both settling 2/5. [a]Attitude PD zeroed, airborne spin-down damping kept. [b]Levelling integrator off, height split kept; the lone 15 cm traversal gives no std.

Contact-loss recovery. ACC settles from every drop and ledge condition tested. The two flight ablations are more nuanced than the design rationale suggests: removing the flight *attitude* PD is a real but modest loss (one fall in ten), while removing the airborne wheel-torque override entirely does not degrade either condition and in fact *lowers* peak pitch, because free wheels driven by the pitch-feedback loop are themselves an effective reaction flywheel. The override's benefit appears only at the envelope edge: at 150 cm both configurations settle 2/5, but the override holds peak-pitch spread to ±2.1° against ±35.0°, so it **bounds landing-attitude variance near the limit** rather than enabling the recovery.

Per-leg terrain adaptation. The criterion is traversal, not survival. Under it, with the straddle gating of §4.6, the robot traverses 10/10 at every height at a straddle roll that grows from 2.4° to 3.9°. The growth is *tracking*, not geometry: mid-straddle (19) commands a leg-length difference of 0.193 m against the 0.20 m step, so ε=7 mm goes unspanned—$\arctan(\varepsilon/w_{\text{track}})$=1.4°—and the remaining 2.5° is servo error at the shortest gravitational moment arm the PID gains were tuned for. Suppressing heading hold through the straddle instead—the behaviour before the roll band was made split-dependent—leaves the mount transient undamped, the yaw error runs to the 0.30 rad leash, and the elevated wheel leaves the slab 1.3 m in. The component ablations are unambiguous. Replacing the split with a symmetric posture falls at *every* curb height; the straddle gating is inert here, as it must be, being driven by the commanded split. Disabling the levelling trim alone holds the 10 cm curb but collapses beyond it: the feedforward table does not span a step this large, and its residual grows faster than the straddle allowance absorbs.

5.4. *Comparison with Classical Baselines and a Preliminary RL Reference*

To establish that ACC is not merely a tuned linear controller we compare six classical baselines under identical environment setups: four derived from the 4-state TWIP model (Grasser et al., 2002) (LQR with decoupled roll/yaw PD through a PID servo layer of ∼0.25 s lag; the same plus a pitch-error integral with conditional anti-windup; a direct-torque variant re-optimised for the zero-lag plant; and PI+AW, adding a conditional forward-position integral) and two 6-state coupled variants adding the roll states $(\phi, \dot{\phi})$, one through the servo and one commanding torque directly. All weights follow Bryson's rule (Bryson & Ho, 1975). A model-free PPO run (5.4M steps, single seed, no domain randomisation or hyperparameter search) is reported alongside them as a difficulty probe rather than a tuned RL baseline (Limitation 4).

Table 5. Classical baselines and the preliminary PPO reference vs. ACC, clean sensors. All rows at 100 Hz control except the PPO reference, which is reported at the 50 Hz rate its policy was trained at.

Configuration	Surv. (s)	θ_{RMS} (°)	ϕ_{RMS} (°)	H_{RMS} (mm)	τ_{RMS} (Nm)
LQR, TWIP (N=20)	0.32	0.006	24.2	205.8	21.99
LQR + AW (N=20)	0.32	0.006	24.2	205.8	21.99
LQR, DT (re-opt.) (N=20)	0.60	0.033	27.5	119.2	8.76
6-St. LQR (N=20)	0.32	0.006	24.2	205.8	21.99
PI + AW, DT (N=20)	0.52	0.181	26.1	171.3	10.42
6-St. LQR, DT (N=20)	0.16	0.001	21.2	128.5	14.44
PPO ref. (50 Hz) (N=20)	3.20	4.82	3.15	54.2	18.40
ACC (N=10)	**20.00**	**0.060**	**0.010**	**0.18**	**3.28**

LQR / LQR+AW / 6-St. use the PID servo; the DT rows command torque directly. The three servo rows are identical because that servo's response dominates before the gain differential between them can act. Survival time is deterministic on all six classical rows (±0.00 s at N=20, seeds {0, 42, 123}). Raw output `outputs/rate_matched_baselines/results.json`.

All six classical baselines fall within 0.60 s. Removing the PID servo path cuts torque 2.5× and nearly doubles survival but *worsens* roll: the recovered authority is spent in the sagittal plane the model describes, not in the lateral one that ends the episode. A negative result is only as strong as its tuning, so the family is swept rather than tuned once: a 46-configuration retuning sweep leaves the outcome unchanged, as does sweeping the coupled model's one empirical constant, the hip-roll-to-roll-acceleration coupling $b_{r,h}$, over an 80× range: the servo variant falls at 0.320 s for every value, with roll RMS spread 0.06 %, even though the re-derived roll gain varies 32×. That is invariance to the parameter rather than insensitivity of the metric, and its mechanism is saturation—the roll channel is commanded against the ±0.7 rad hip-roll limit throughout the episode at every value tested. The preliminary PPO reference stands longer at a single height but falls in 85 % of multi-height command transitions—the observation the structured prior is meant to address.

What a 100 % *fall rate does and does not claim.* LQR balances wheeled bipeds on real hardware—Ascento (Klemm et al., 2019) and DIABLO (D. Liu et al., 2024) both do it—so a table in which every LQR variant falls invites the reading that the plant or the tuning is at fault rather than the method. The answer is in which axis ends the episodes. None of these controllers fails at the task LQR is credited with on those platforms: the servo variants hold pitch to 0.006° RMS, and the full-state design of Section 5.4.1 holds 0.12° for the full 20 s. What terminates every episode is *roll*, on a morphology carrying two lateral degrees of freedom per leg that those platforms do not have—and the published designs do not ask a single LQR to close that axis either, DIABLO running separate yaw, split-angle, height and roll loops around its balance controller (D. Liu et al., 2024). The claim these rows support is therefore narrow and morphological: reduced-order LQR of the form the wheeled-biped literature uses does not transfer to a 10-DOF articulated-leg plant *as a single controller for all axes*. It is not a claim that LQR cannot balance a wheeled biped, and Section 5.4.1 is the control experiment that keeps the distinction from resting on our tuning.

The sharper form of the objection is internal to this paper: ACC's own ablation ladder contains a rung, L0, that is proportional-only on the wheel channel—no position term, no integral, no anchor—and it stands 10/10 (Table 1). Why should a hand-tuned proportional law survive where an optimal one does not? Because L0 is not a linear controller for the whole robot: it is ACC with the anchor removed, and it retains the dedicated roll, yaw and posture channels that no row of Table 5 has. The comparison in that table is therefore neither optimal-versus-hand-tuned nor linear-versus-nonlinear; it is one linearisation for all axes against per-axis structure. Remove the structure and add optimality, and the plant falls in 0.32 s; keep the structure and strip everything else back to a proportional term, and it stands—at 17.461 mm repeatability and a 69.7 mm limit cycle, which is the gap the remaining terms buy.

5.4.1. A full-state LQR designed on the plant itself

All six baselines are designed on a reduced model, so their failure is open to the objection that the model, not the method, is what fails. The control experiment that settles this is a regulator derived from the plant directly: a damped Gauss–Newton solve for a true static equilibrium at the commanded height, central differences on the whole held-input control step of the complete 16-DOF, 10-actuator MuJoCo model for the transition Jacobians, and a discrete-time Riccati solve on the 30 states left after removing the two cyclic wheel angles, with Bryson's rule fixing $\mathbf{Q}$, $\mathbf{R}$ and a uniform control weight r swept over six decades. The design is sound on every diagnostic that bears on it—residual unbalanced force 0.428 N against 79.46 N of body weight, $\mathbf{A}$ of full rank 30 and PBH-stabilisable, Riccati relative residual $\sim 10^{-14}$, closed-loop max $|\lambda|$=0.9911—and $\kappa(\mathbf{A})\approx 10^{10}$ is a 50 Hz artifact (2.6×10^{7} at 100 Hz) from strongly contracting contact modes that obstructs neither the solve nor the feedback.

What the sweep shows is a slow divergence a short horizon hides. Over 20 s the design looks successful (r=1000 at 0.65 m: 0 % falls, 11.0 mm height RMSE); at 60 s every configuration falls in 100 % of episodes at 22.2 – 57.0 s, with planar drift growing monotonically to 0.43 – 0.63 m—that drift, not attitude, ends them. Nor does one weight cover the band: r=100 stands 20 s at 0.60 m and 0.69 m but falls at 0.65 m, r=1000 does the reverse, and under random height commands the fall rate runs 45 – 80 %; the maximum recoverable push is 10.0 – 20.2 N against ACC's $F_{\min}$=82 N. Full-state optimal linear feedback about a single equilibrium therefore buys two orders of magnitude of survival over the reduced-order designs and still does not close the problem. That reading rests on a horizon argument, which only carries if the proposed controller is held to the same test; it was. Extending ACC's quiet-stance window by the same factors returns the opposite verdict—10/10 at 300 s, 4/4 at 1800 s, residual drift 0.0004 mm/min against this design's 0.43 – 0.63 m before falling (Section 5.1.4).

5.5. Whole-Body Control Baselines

Whether a tuned whole-body controller could match ACC was tested over 225 scenarios spanning fixed-height balance, height transitions and push recovery at 20 – 120 N under cloned initial conditions (481 000 control steps, no gain retuning): a task-priority quadratic-program WBC—minimising CoM height error, torso orientation error, posture damping and wheel regularisation subject to full-body dynamics and contact constraints—used as the *primary* balance controller (W1), and the same WBC blended as an *assist* on ACC via $\boldsymbol{\tau}_{\text{assist}} = \boldsymbol{\tau}_{\text{ACC}} + \alpha(\boldsymbol{\tau}_{\text{wbc}} - \boldsymbol{\tau}_{\text{ACC}})$ (W2). So that the comparison is not a verdict on our own implementation, the WBC was verified beforehand against executable references: its analytic contact Jacobian against MuJoCo's `mj_jac`, its task stack against four independently selectable weightings, and its height reference against the CoM-z convention of §3.1.

W1. The standalone QP-WBC falls in **225 of 225** scenarios, where ACC falls in none; it holds the robot upright for 0.570 s and spends 97.3 % of every episode fallen, and the failure is *lateral* (roll RMS 93.45° against ACC's 3.71°). Neither implementation quality nor task weighting decides it: a degraded build solving only 85.85 % of steps falls at 0.52 s against the verified build's 0.570 s at 99.99 % solved, and four task-weight modes move the fallen fraction by 0.1 percentage points. The mechanism is that the posture task is a velocity damper, not a position tracker—its reference is rebuilt from the measured joint vector every control step, so it collapses to $-k_{d,\text{posture}}\dot{\mathbf{q}}$ and can slow a postural drift but not reverse one.

W2. The two assist configurations together constitute a gate ablation. With the authority gate *disabled* (α=0.25 fixed) the assist arm never falls—ACC carries it—but degrades ACC on every metric but one: pitch RMS from 0.52° to 1.295°, yaw drift by 1.9×, planar drift by 2.0×. With the gate *enabled*, the admitted authority on the one variant where it opens at all is $\bar{\alpha}$=0.0061, 2.5 % of the permitted maximum, and there the gated assist agrees with ACC alone to four significant figures; on the remaining four variants α is identically zero. Roll is the single exception: with the gate disabled it improves systematically from 3.71° to 3.20°, because ACC regulates the lateral axis with decentralised hip-roll PD whereas the WBC uses a centralised orientation task exploiting full mass-matrix coupling. That 18 % of roll costs 2.5× the pitch error and 2.0× the drift, so a centralised lateral task is a credible direction for ACC's weakest axis—the 98.2 N mean lateral bearing of Sec. 5.2—but not through torque-level blending with a posture stack that cannot hold a posture. Compute is not the constraint: the QP solve costs 0.169 ms mean and 0.30 ms p99 (OSQP (Stellato et al., 2020)).

5.6. Architectural Analysis

Robustness to sensor noise and actuator delay. A factorial sweep over four noise levels and three delay levels (N=20 per cell, 240 trials) bounds ACC on both axes at once (Table 6). One control period of delay is benign for stability and expensive for performance: at 10 ms there are zero falls on clean and low noise, but $F_{\max}$ nearly halves and idle RMS rises 5.6×. Three control periods is a different regime: at 30 ms the platform falls in 20 of 20 trials on *clean* sensing. Noise alone is survivable up to the medium level; what ACC does not tolerate is the *conjunction*: medium noise with 10 ms of delay produces the only sub-catastrophic falls anywhere in the grid. High noise is unrecoverable at every delay including zero, because the asymmetric envelope cannot distinguish noise from motion—the same mechanism that produces the sharpest single sensitivity in the design, idle repeatability degrading 71-fold under medium noise at zero delay as the envelope gate releases and re-engages the integral. ACC is therefore specified against a filtered velocity estimate, a requirement on the state estimator rather than a limit on the controller.

Resolving the delay axis. The grid quantises delay to the control period, so the knee was located separately with the transport delay applied per *physics* substep (2 ms at 500 Hz), the clean-sensing harness being deterministic. The knee lies between 6 ms and 8 ms: $F_{\max}$ is flat to within 2.4 % out to 6 ms (93.2 – 95.5 N), then collapses 45 % to 51.0 N and declines monotonically to the 10 N search floor by 20 ms—a genuine loss of margin rather than a narrow phase-alignment notch, and the 6.6 – 8 ms end-to-end budget estimated in §6 lands inside the 6 – 8 ms bracket rather than clear of it. Losing the push margin and losing the platform are different thresholds: on quiet stance the standing threshold is 14 – 16 ms,

Table 6. Robustness to sensor noise and actuator delay, N=20 per cell. Falls: trials terminated by a fall, out of N. Idle RMS: lateral (world x) CoM std. over 20 s after a 3 s settle, *surviving trials only*. $F_{\max}$: binary-search lateral (world $+x$) push, ±5 N tolerance, search floor 10 N. Delay is quantised to the control period here; it is resolved five times more finely in the text.

Condition	Falls	Idle RMS (mm)		$F_{\max}$ (N)	
		Mean	±Std	Mean	±Std
Clean, 0 ms delay	0/20	0.43	0.06	95.7	0.8
Clean, 10 ms delay	0/20	2.43	0.21	50.5	7.9
Clean, 30 ms delay	**20/20**	— (all fell)		—	
Low noise, 0 ms delay	0/20	5.86	2.53	96.0	1.0
Low noise, 10 ms delay	0/20	10.66	3.70	51.3	8.9
Low noise, 30 ms delay	**20/20**	— (all fell)		—	
Med noise, 0 ms delay	0/20	30.82	12.13	93.5	5.2
Med noise, 10 ms delay	**2/20**	35.38	21.63	51.3	13.4
Med noise, 30 ms delay	**20/20**	— (all fell)		—	
High noise, all three delays	20/20	— (all fell)		—	

Noise (1-σ): low = 0.01 rad/s angular velocity, 0.01 m/s^2 gravity, 0.001 rad joint position; med = 5× low; high = 10× low. Delay is a transport lag on the actuator command. High noise fails identically at all three delays, so those rows are merged. Because the idle column averages survivors only, the medium-noise 10 ms row is mildly optimistic and its falls column is the primary metric. Produced by `scripts/collect_robustness_sweep.py`; raw output `outputs/paper_statistics/robustness_sweep.json`.

station held through 14 ms at 10.84 mm idle lateral RMS against 0.43 mm undelayed.

The sweep also answers whether that knee is architectural. Holding the delay at 8 ms and sweeping one profile constant at a time gives not a gently sloped optimum but a sharp *local dip*: the shipped value of each of four constants tested scores 51.0 N while nearly every neighbouring value recovers 78 – 93 N, which is not how an exhausted margin behaves. Carrying the largest single-knob change forward—sagittal velocity damping doubled to 45 Nm/(m/s), denoted ACC-DH—leaves the controller identical at 0 – 2 ms and moves the knee outward to 8 – 10 ms (51.0 → 93.2 N at 8 ms) for +1.4 % idle lateral RMS. We report ACC-DH as an evaluation variant and do *not* adopt it, since the constant it changes also governs driving acceleration and settling.

Gate sensitivity and rate-limit invariance. Finite-difference gradients reveal a stark hierarchy: the release coefficient α_r dominates (sensitivity 220.4, 20× the next and 130× the pitch gates), while the pitch thresholds are effectively insensitive safety interlocks. The three gate failure modes all map onto measured ablation rows of Table 3: stuck-at-1 causes windup and a fall (S4), stuck-at-0 yields the P-only limit cycle (L0), and stuck-half-open triggers the parametric resonance of S5. The 400 Nm/s slew limit is not what sets the push ceiling: over an 8× span lateral $F_{\max}$ is 92.5±1.1 N at 100 Nm/s and 94.4±0.0 N at 200, 400 and 800 Nm/s, while during a 90 N push the commanded slew saturates exactly at each bound—so the ceiling is architectural rather than actuator-limited.

Compound disturbance. Because the posture map is height-scheduled, a commanded transition and a push compete for the same leg-torque budget, so the standard lateral impulse is applied at the midpoint of a 1 s commanded height ramp of size Δh about the 0.404 m nominal, with Δh=0 as the matched static control (N=10 per row). Inside the calibrated band a simultaneous height command is nearly free (a +5 cm rise costs 5.1 % of push margin; a −5 cm squat is *better* than static, +6.5 %), and the degradation outside it is one-sided: squatting −10 cm costs nothing measurable (p=0.99) while rising +10 cm costs 68.7 %, because rising drives the knee toward extension—where the leg Jacobian is worst conditioned—while raising the CoM. Re-parameterising the *same* posture family so achieved height matches the command recovers 78 % (p=1.5×10^{-4}) but closes only 35 % of the deficit. Sequential disturbance is absorbed cleanly: a 90 N lateral impulse followed 2 s later, before the first ringdown has decayed, by a 60 N counter-directed impulse is survived **10/10** at episode peak pitch 11.89±0.51°, almost all of it from the first impulse—what τ_{release}≈1.5 s is designed to produce, the damping boost still partly engaged when the second impulse arrives.

5.7. Local Stability Certificate at the Standing Fixed Point

ACC's gates make it a switched system, of the kind partitioned by discrete switching surfaces in the hybrid-systems literature (Liberzon, 2003) and softened by hysteresis in supervisory switching logic (Hespanha et al., 2003). Everything reported so far is behavioral: the platform does not fall, for 30 min (Section 5.1.4) and under every disturbance of Section 5.2. That is evidence, not a stability statement. This section supplies one, in three parts—where the switching is, what the one live switch does, and what the linearized closed loop is—and is explicit about what it does not cover.

At quiet stance the switching is inactive, by a measured *margin.* Over the last 20 s of a 60 s settle, 19 of the 21 gates the controller exports sit at exactly 0 or exactly 1 at every control step: bit-exact saturation, not approximate. Two anchor gates are not exported and are reconstructed from controller state. The proximity gate's argument $|e_{\text{sag}}|$ peaks at 24.8 mm against its 50 mm knee (16.1 – 33.2 mm across the five height variants, a 1.5–3.1× margin), and the envelope gate's argument peaks at 1.5×10^{-7} m/s against its 0.25 m/s knee, a margin of $1.7{\times}10^{6}$. Two gates do sit inside their bands—the anti-twist gate at ≈0.53 and the centering gate at ≈0.986—but a smoothstep is C^{∞} in the interior of its band, so those are ordinary smooth feedback rather than switching. In a neighbourhood of the fixed point every branch selection in the controller is therefore frozen, the closed loop is a single smooth map, and Lyapunov's indirect method applies. Chattering is not avoided here *because* the gates are C^1; it is avoided because the operating point stands far from every switching surface, and the margins above are how far.

The one switch that does fire is a contraction on both branches. The asymmetric envelope of Eq. (14) selects $\alpha \in \{0.35, 0.0067\}$ on the sign of $|v_{\text{sag}}| - \text{EMA}_{t-1}$, and in quiet stance that sign flips at essentially every step, so the composition is only C^0 *in time*. For a fixed input level the tracking error $e_t = \text{EMA}_t - |v_{\text{sag}}|$ obeys $e_{t+1} = (1 - \alpha_t)e_t$, so $|e_{t+1}| \leq 0.9933\,|e_t|$ under *either* branch. Hence $V(e) = e^2$ is a common Lyapunov function for the switched pair, and the envelope is globally exponentially stable under arbitrary switching, worst-case rate set by the slow branch (τ=1.49 s). The classical obstacle—that switching among individually stable subsystems can destabilize the whole (Liberzon, 2003)—cannot arise, and no dwell-time or hysteresis condition is required (Hespanha et al., 2003). The asymmetry buys the attack/release separation the design wants; it does not have to buy stability. The C^0 property is discharged twice over: the discontinuity lives in a provably contracting scalar subsystem, and it reaches the loop only through a gate that is flat (≡ 1) across the entire quiet band, so its switching cannot modulate loop gain in the certified region.

The linearized closed loop. We central-difference the whole 100 Hz control step—control law plus five physics substeps—about the settled state, over an augmented vector of 57 coordinates: 30 plant (16 velocities and 14 configuration tangents; the two wheel angles are cyclic and removed, as in Section 5.4.1), 3 of estimator memory, and 24 of controller memory. The adaptive-bias trim is held at its converged value—it is an 800-tap moving average, memory rather than dynamics, and freezing it makes the one-step map time-invariant instead of 300-periodic. Table 7 reports the result: at all five heights the spectrum has exactly two eigenvalues on the unit circle, none outside it, and a reduced spectral radius of 0.997 87 – 0.998 18, i.e. a slowest stable mode of 4.7 – 5.5 s. The classification is not a threshold judgement—the gap from the unit circle down to the next mode is $1.8{\times}10^{-3}$, four orders of magnitude above the 10^{-7} sensitivity of the spectrum to the differencing step.

Both centre directions are identified, and both are bounded by measurement. The first is lateral translation: a differential-drive base produces no lateral force to first order, so this is a non-holonomic centre direction and appears at $|\lambda|$=1 to 10^{-12}. The second is body yaw, and it is deliberate: the heading channel carries an explicit deadband, $\text{ss}(|e_{\text{yaw}}|; 0.02\,\text{rad}, 0.08\,\text{rad})$, measured at exactly 0 here, so absolute yaw enters no control law while the error stays inside ±1.15°. Neither is left to the linearization. Perturbing each coordinate at the fixed point and running 300 s of the full nonlinear simulator, a 1.000 mm lateral offset returns 0.997 mm—no growth, late rate $-1.1{\times}10^{-5}$ mm/s—and a 0.500° yaw offset creeps to 0.565° at $+4.8{\times}10^{-5}$ °/s, which needs 3.3 h to reach the deadband edge at which heading control re-engages. The lateral mode is the same one the 30 min run of Section 5.1.4 measures as 0.14 mm/hour of creep; the certificate and the long run are measuring the same thing from two directions.

Cross-validation, and scope. The linearization is checked against the plant it was taken from. Random plant perturbations at 10^{-4}, 10^{-3} and 10^{-2} decay in the full nonlinear simulator—adaptive bias live, nothing frozen—at ρ=0.99521±0.00237, 0.99513±0.00117 and 0.99547±0.00355, against the 0.995 30 predicted for the anchor-integral mode (τ=2.12 s, 0.033 Hz), which is the mode a generic perturbation excites: agreement to 2×10^{-4} over two decades of amplitude. What this does *not* establish is equally definite. The certificate is local and on the nominal model, at zero delay and clean sensing; it estimates no region of attraction, and the push results of Section 5.2 lie outside it by construction, since a push is precisely what drives the gates off their flat regions—that is what the gates are for. What it does establish is that the point ACC parks at is exponentially attracting in 55 of 57 directions and neutral in the other two, with those two named, mechanistically explained, and separately bounded.

Table 7. Local stability certificate at the standing fixed point, across the five calibrated height variants. Central-difference linearization of the 100 Hz closed-loop step after a 60 s settle; "Flat" counts exported gates that are bit-exactly saturated over the last 20 s. ρ_{red} excludes the two centre directions (lateral translation and deadbanded yaw), which are bounded separately by 300 s nonlinear probes.

Variant	CoM z (m)	Flat	Ctr.	Unst.	ρ_{red}	τ_{slow} (s)
Nominal	0.4069	19/21	2	0	0.99818	5.49
High small	0.4164	18/21	2	0	0.99787	4.70
High tiny	0.4128	19/21	2	0	0.99817	5.46
Low small	0.3973	19/21	2	0	0.99813	5.34
Low tiny	0.4009	19/21	2	0	0.99815	5.41

Produced by `scripts/acc_stability_certificate.py`; full spectra, gate traces and probe trajectories in `outputs/paper_verification/acc_stability_certificate.json`. Spectra are insensitive to the differencing step at the 10^{-7} level.

6. Discussion and Conclusion

Every gate in ACC was added after tracing a measured failure mode to its root cause: a bistable symmetric envelope, global integral windup, parametric pumping of a scheduled stiffness, hard-leash oscillation and self-referential stiffness. All thresholds were frozen before the ablation sweep, so Table 3 scores a fixed specification rather than a tuned one, and because the gating variables are physical quantities the controller runs without training and is hypothesised, pending hardware validation, to carry a reduced sim-to-real gap (Grimminger et al., 2020).

Statistical power. Welch's t-test at Bonferroni-adjusted α=0.007 (7 comparisons) makes L0→L1 and L2.5→L3 significant (p<0.001; Cohen's d=5.4 and 4.1). Flight and terrain results are N=10 with Clopper–Pearson 95% CI [0.69, 1.00]; push ablations resolve ~2 N, and no claim rests on less.

Limitations. (1) *Simulation only.* All results are MuJoCo; hardware validation remains future work. (2) *Deployment timing is the binding constraint.* ACC as tuned requires a 100 Hz loop and a latency clearing the 6 – 8 ms push-margin cliff of §5.6. Rate independence is not claimed: at 50 Hz with recomputed coefficients the platform does not stand, and the cause—plant-mode excitation or phase lag at Δt=20 ms—is unresolved. (3) *Two calibrated envelopes are non-uniform.* Oblique ledge descent succeeds only within the stagger $w_{\mathrm{track}} \tan\phi/v$ that the leg split can absorb, giving two regimes split by a dead band (192 trials at 30°: 100 % to 15 cm, closed over 22 – 27 cm, reopening past 30 cm); and a +10 cm height command during a push costs 69 % of $F_{\max}$, descent being free at every amplitude. (4) *The learning comparison is preliminary, and one classical family remains untested.* The model-free PPO run (5.4M steps, one seed, no domain randomisation (Tobin et al., 2017) or hyperparameter search) is a difficulty probe bounding nothing about RL here; the comparison this controller is built for is residual learning *over* the prior rather than against it, and that comparison is not run here. A tuned convex MPC over the linearisation of §5.4.1 is the natural next comparator. (5) *Generality across morphologies is undemonstrated.* Every result is for one 10-DOF platform with one set of gains.

Deployment risks. ACC needs only raw torque, a 400 Nm/s rate limit, an IMU and encoders. The binding figure is the 6 – 8 ms latency cliff in push margin (§5.6); the stability threshold is roughly twice as far out (station held through 14 ms), the ordering the architecture predicts. An estimated non-RTOS budget totals ≈8 ms: sensor readout ~1 ms, state estimation ~2 ms, actuation ~3.5 ms—all assumed—plus a *measured* ≤1.5 ms for the control step, reported on the same footing as the discarded WBC baseline's QP solve time and replacing a ~3 ms assumption (1.27 – 1.52 ms p99 across five repeats of 5000 steps at the quiet-stance operating point; the CoM estimate inside it is already charged to the state-estimation line, so that work is counted twice).

Host wall-clock is nevertheless the wrong quantity to extrapolate to an embedded target, and the spread above is why. The reproducible quantity is the operation count: 27 664 floating-point operations and 159 transcendentals per step over 1001 scalars, identical on every repeat, with no iterative solve, no matrix factorization and no data-dependent control flow—every gate of §4.3 is a branchless select—so worst-case execution time equals average-case, which a QP-based controller cannot offer. A 480 MHz Cortex-M7 retires that in ~0.07 ms and a 100 MHz Cortex-M4 in 0.34 ms; compute reaches the 1.5 ms budgeted above only below about 23 MHz. It is thus the one budget line that shrinks on the target rather than growing, since ~95 % of the measured 1.2 ms sits outside the compiled kernel—CoM and support-polygon work plus Python dispatch, which a C port deletes or reduces to a few hundred operations of forward kinematics. Charged at 0.1 ms the estimate is 6.6 ms. Against the 6 – 8 ms cliff the instrumented figure sits at the top of the bracket and the projected one at its bottom, so what decides the matter is readout and actuation, not compute. Two mitigations move the loop clear and compose: a real-time OS recovers 2 – 4 ms (loop at 2.6 – 4.6 ms projected), and doubling the sagittal velocity damping moves the knee to 8 – 10 ms without touching the schedule. Real-time scheduling remains the recommendation; hardware repeatability is projected at 1 – 3 mm rather than 0.294 mm.[1]

Conclusion. We presented ACC, a two-channel torque assembly in which a PD balance core is augmented by a proximity-gated anchor integral, achieving 0.294±0.015 mm idle repeatability on the sagittal axis—a 59× improvement over the proportional-only baseline (N=10, Table 1)—together with omnidirectional push survival from $F_{\min}$=82 N to a 107 N median. Absolute accuracy is reported separately: the robot parks 27.0 mm from commanded home and holds that standoff to 0.009 mm, which a closed-form model predicts as the static price of a bounded integrator memory to within 0.45 %. Six classical baselines and a full-state LQR derived from the plant itself all fall, and integral action does not prevent it. ACC tolerates moderate noise, is insensitive to the slew bound over an 8× span, and handles contact loss and terrain adaptation without training; its sharpest deployment constraint is actuator latency. The contribution is a *structured classical prior*: an interpretable controller whose gating encodes the plant's measured failure modes. Everything claimed for it is a directly measured property of the controller standing alone—the mechanism decomposition, the chirality of the push envelope, the closed-form parking offset and the stability certificate of Section 5.7—so the results do not wait on the residual stack the prior was built to serve. That stack is one downstream use among the future work, alongside hardware validation, a resourced learning comparison, and conditional activation for locomotion.

Code and Data Availability

The controller, MuJoCo model, evaluation scripts and raw result files are at `https://github.com/thuong180702/Wheeled-bipedal-robot-simulation` (branch `main`); each table's note names its script and result file. The extended version there carries the full derivations, WBC details and parking-offset model.

[1] Produced by `scripts/measure_control_step_cost.py`; raw output `outputs/paper_verification/control_step_cost.json`.

7. Notes on contributors

Van Thuong Nguyen is an independent researcher based in Viet Nam, working on legged and wheeled mobile robots. His research interests include classical and learning-based control for wheeled bipedal platforms, whole-body control, and structured model-based controllers used as action priors for residual reinforcement learning.

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